SANZ WADIBHASME

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LinkedIn ↗ | GitHub ↗ | Portfolio ↗

PROFESSIONAL SUMMARY

B.Tech student in AI & Data Science with hands-on experience in Machine Learning, Deep Learning, Generative AI, RAG, Computer Vision, and Data Analytics. Skilled in building and deploying AI-driven applications using Python, FastAPI, React, Streamlit, vector databases, and modern ML frameworks and seeking opportunities in AI/ML, Data Science, and AI Engineering.

EDUCATION

B.Tech. — Artificial Intelligence & Data Science Expected Apr 2027
Faculty of Engineering and Technology, AIIHER (DU), Sawangi, Wardha | CGPA: 8.0/10

EXPERIENCE

AI Intern — *Fireblaze Technologies Pvt. Ltd.* | Apr 2026 – Sept 2026

- Developed and contributed to Fireblaze GPT, Adaptive AI Learning Companion, Smart Surveillance & Safety Monitoring, and RoadShield AI across AI/ML development and integration.
- Built RAG pipelines, computer vision models, and Fast API backends, using vector databases and open-source AI tools.
- Built Google Ads, GA4 & GSC dashboards in Looker Studio for performance analytics.

Web Development Intern — Cosmos Digital | Apr 2025 – Jun 2025

- Developed responsive frontend interfaces and dashboard components using React.js, HTML, CSS, and JavaScript.

SELECTED PROJECTS

RoadShield AI — Autonomous Road Hazard Detection & Risk Assessment. 🔗

- Built a YOLOv8-based system to detect 7 road hazards including potholes, manholes, cracks, debris, and speed bumps using a 25K+ image dataset, with risk scoring, tracking, GPS mapping, and Firebase/Google Maps integration.

NeuroPulse AI — Neuromorphic Healthcare Platform 🔗

- Built a software-simulated neuromorphic healthcare platform for multi-modal biosignal analysis using Spiking Neural Networks (SNNs) across ECG, EEG, and EMG, with spike-attribution explainability, energy-efficiency analysis, and an interactive clinical decision-support dashboard.

Autonomous Satellite Mission Debrief & Anomaly Intelligence System 🔗

- Built an AI-driven satellite telemetry analysis system to detect anomalies and generate mission debriefs using RAG, knowledge graph reasoning, and local LLMs, with automated retrieval of relevant mission information.

TECHNICAL SKILLS

Languages: Python, Java, SQL
Data Science & Analytics: Data Analysis, EDA, Data Preprocessing, Feature Engineering, Predictive Modelling, Data Visualization, Matplotlib, Seaborn, Grad-CAM
AI / ML: Machine Learning, Deep Learning, NLP, Computer Vision
Generative AI: RAG, Embeddings, Vector Search, LLM/SLM Applications, Prompt Engineering
Frameworks: PyTorch, TensorFlow, Scikit-learn, FastAPI, Streamlit, React
Databases / Tools: ChromaDB, GitHub, Docker, Google Analytics 4, Looker Studio

RESEARCH & PUBLICATIONS

- Binary and Multi-Label Ocular Disease Detection from Retinal Fundus Images Using MobileNetv2 and ResNet50 with Grad-CAM Interpretability and Clinical Web Deployment - *IEEE, ICCVBIC 2026* | IEEE Xplore ↗
- Semantic Consistency Drift Monitoring for Training-Free Adversarial Prompt Injection Detection in Agentic LLM Pipelines - *IEEE, ICCVBIC 2026* | IEEE Xplore ↗

ACHIEVEMENTS & LEADERSHIP

- Academic Achievements: Received a Letter of Recognition for securing 2nd Rank among the entire 1st Year; awarded a Trophy & Certificate for 3rd Rank in the AI & Data Science Department in both 2nd and 3rd Year.
- GDG/GDSC Member: Completed Google Cloud Study Jam & GenAI cohorts, earning Google swag.
- Publicity & Sponsorship Lead - C4GT (Code for GovTech) Club.
- Anchored events at IGNITRA and IEEE conferences, supporting event coordination and audience engagement.

CERTIFICATIONS

- Data Mining - NPTEL (2026) ↗
- Machine Learning with Python - IBM (2025) ↗
- Intel Unnati Industrial Training – 2025 ↗
- Databases and SQL for Data Science with Python - IBM (2025) ↗